

2025-2026 – Econ 0016 – International Macroeconomics

Chapter 2: Current Account Sustainability

Schmitt-Grohé, Uribe and Woodford, “International Macroeconomics: A Modern Approach”

Franck Portier

f.portier@ucl.ac.uk

University College London

Version 1.0

09/01/2026

0. Introduction

- ▶ Natural question from our description of the recent history of the U.K. and U.S. external accounts:
- ▶ Are the observed trade and current account deficits sustainable in the long run?
- ▶ Here we develop a framework to address this question.

1. Can a Country Run a Perpetual Trade Balance Deficit?

- ▶ It depends on whether the country is a net debtor or a net creditor.
- ▶ If it is a net debtor, that is, if its net international investment position (NIIP) is negative, then the answer is no.
- ▶ Why? Because in this case, the country will have to run a trade balance surplus at some point to service its debt.
- ▶ If the country is a net creditor to the rest of the world, that is, if its net international investment position is positive, then it can run a perpetual trade deficit and finance it with the interest generated by its net investments abroad.
- ▶ Let's analyze this issue more formally.

1. Can a Country Run a Perpetual Trade Balance Deficit?

- ▶ Consider an economy that lasts for two periods.
- ▶ It starts period 1 with NIIP of B_0 and r is the constant interest rate.
- ▶ Then, the country's net investment income in period 1 is given by

$$\text{Net investment income in period 1} = rB_0.$$

- ▶ Trade balance in period 1: TB_1 .
- ▶ The country's NIIP at the end of period 1 is¹

$$B_1 = B_0 + CA_1 = B_0 + TB_1 + rB_0 = (1 + r)B_0 + TB_1 \quad (1)$$

- ▶ and for period 2:

$$B_2 = B_1 + CA_2 = B_1 + TB_2 + rB_1 = (1 + r)B_1 + TB_2 \quad (2)$$

- ▶ At the end of period 2, the country cannot hold assets or debts because no one will be alive in period 3 to collect (the world ends in period 2). This means that

$$B_2 = 0. \quad (\text{"transversality condition"}) \quad (3)$$

¹We assume that the income balance is equal to net investment income and that net unilateral transfers and valuation changes are zero, so that $CA_t = TB_t + rB_{t-1}$.

1. Can a Country Run a Perpetual Trade Balance Deficit?

- ▶ Combining (1), (2), and (3):

$$(1 + r)B_0 = -TB_1 - \frac{TB_2}{(1 + r)}, \quad (4)$$

- ▶ Initial NIIP including interest, $(1 + r)B_0$, equals the present discounted value of trade deficits.
- ▶ From this expression:
 - × if the country starts out as a net debtor, $B_0 < 0$, then it must run a trade balance surplus at some point,
 - × if the country starts out as a net creditor, $B_0 > 0$, then it can afford running trade deficits in both periods.
- ▶ This result holds not just for two-period economies, but for economies lasting any number of periods, including an infinite number of periods.
- ▶ Example: Since the United Kingdom is a net debtor, the present analysis implies that it will have to revert its trade balance deficits at some point in the future.

2. Can a Country Run a Perpetual Current Account Deficit?

- ▶ The answer is, again, yes ...
- ▶ .. provided the country's initial net international investment position is positive.
- ▶ Recall that the change in the NIIP is the current account

$$CA_1 = B_1 - B_0.$$

- ▶ Similarly, in period 2 we have

$$CA_2 = B_2 - B_1.$$

- ▶ Combine these two expressions to eliminate B_1 and recall that $B_2 = 0$ to obtain:

$$B_0 = -CA_1 - CA_2$$

↪ the country can run current account deficits in both periods only if the initial NIIP is positive.

- ▶ This result holds for economies lasting any finite number of periods.

3. Saving, Investment and the Current Account

Gap between Saving and Investment

- In any period t , saving, investment, and the current account are linked by the identity

$$CA_t = S_t - I_t.$$

- The country's saving in excess of what is needed to finance domestic investment must be allocated to purchases of foreign assets (*Where else could those resources go?*).
- The change in the net foreign asset position is precisely the current account.
- How to derive that equation?

$$\underbrace{Q_t + IM_t}_{\text{supply}} = \underbrace{C_t + G_t + I_t + X_t}_{\text{demand}}$$

with gross domestic product Q_t , imports IM_t , private consumption C_t , government consumption G_t , investment I_t and exports, X_t .

3. Saving, Investment and the Current Account

Gap between Saving and Investment

$$\underbrace{Q_t + IM_t}_{\text{supply}} = \underbrace{C_t + G_t + I_t + X_t}_{\text{demand}}$$

- Add net investment income, rB_{t-1} to both sides of the previous expression and recall that the trade balance is the $TB_t = X_t - IM_t$:

$$\underbrace{Q_t + rB_{t-1}}_{\text{GNP}_t} = C_t + G_t + I_t + TB_t + rB_{t-1}$$

- National income (a.k.a. Gross National Product, GNP) is $Y_t = Q_t + rB_{t-1}$
- We also have

$$CA_t = rB_{t-1} + TB_t.$$

so that

$$Y_t = C_t + G_t + I_t + CA_t \quad (5)$$

3. Saving, Investment and the Current Account

Gap between Saving and Investment

- Finally, the difference between national income and private and public consumption is national saving, or

$$S_t = Y_t - C_t - G_t.$$

- Rewrite (5) as

$$CA_t = \underbrace{Y_t - C_t - G_t}_{S_t} - I_t,$$

we get the expression we were looking for:

$$CA_t = S_t - I_t,$$

which says that the current account is the gap between saving and investment.

3. Saving, Investment and the Current Account

Gap between National Income and Domestic Absorption

- ▶ Domestic absorption A_t is defined as the sum of private consumption, government consumption, and investment:

$$A_t = C_t + G_t + I_t.$$

- ▶ Combining this expression with (5), we can express the current account as

$$CA_t = Y_t - A_t,$$

which states that the current account is the gap between national income and the domestic absorption of goods and services.

4. Summing Up

- a. A country that is a net external debtor cannot run a perpetual trade balance deficit.
- b. A country that is a net external debtor cannot run a perpetual deficit in the current account. This result applies to economies that last for any finite number of periods.²
- c. The current account can be expressed as the gap between saving and investment, $CA_t = S_t - I_t$, as the gap between national income and domestic absorption, $CA_t = Y_t - A_t$, as the change in the country's net foreign asset position, $CA_t = B_t - B_{t-1}$, or as the sum of net investment income and the trade balance, $CA_t = rB_{t-1} + TB_t$.
- d. All four expressions above are identities that must hold at any time in any country.

²Remark: for infinite horizon economies, perpetual current account deficits are possible even if the country is an external debtor, if the economy is growing and dedicates a growing amount of resources to pay interest on its external debt.

5. The Road Ahead

- ▶ All four of the above expressions for the current account are accounting identities. They do not provide any explanation, or theory, of the determinants of the current account.
- ▶ To understand what determines the current account we need a model, that is, a story of the economic behavior of households, firms, governments, and foreign residents.
- ▶ This is the focus of the following chapters.

